Training plan: effective strength training

A few principles decide whether you grow, or whether you just show up at the gym. Here they are, in plain language. Right under them is a ready-made frame you can run at home or in a gym. No miracles. Be Effective: do the few things that actually work, and do them regularly. 💪

Progressive overload: the engine of growth

This is principle number one, and everything else hangs off it. A muscle has no reason to get stronger or bigger until you ask it for more than it is used to. When you regularly add a slightly higher demand (more reps, more weight, more sets, cleaner execution), it adapts and moves up. Without a gradual increase in load you stall after a few weeks, however hard you work [1,6]. That is why this whole guide sits on one question: *am I a little further along today than last time?*

Volume: the main lever for muscle growth

If you want to build muscle, the single most important factor is **weekly volume**: how many hard sets you do for a given muscle group in a week. The relationship is dose and response: more sets (within reason) means more growth. A practical target for hypertrophy is about **10–20 sets per muscle group per week** [2]. A beginner does fine at the low end (even 10 sets) and only moves up over time, as tolerance rises. For strength itself, volume helps too, but the slope is less steep than it is for muscle size [7].

Frequency: spread the volume over more days

What matters is less "how many times a week I go to the gym" and more **how often I load each muscle group**. At the same weekly volume, spreading the work to **2 times a week per muscle group** comes out better (or at worst the same) than stuffing it all into one session [3]. That is why a beginner does well on full-body (each muscle group 3 times a week) or upper/lower (each muscle group 2 times a week). You meet the muscle more often, and you do each set in better form.

Load and reps: where muscle grows

Good news: muscle grows across a **wide rep range, roughly 5 to 30 per set**, as long as you take the set close enough to failure. Lighter and heavier weights build a comparable amount of muscle when the work is honest [4]. For **maximum strength** you also need to train with heavy weights and low reps (about 1-6). Strength is largely a

nervous-system skill, and it wants specifically heavy loading [4,6]. For a beginner a sensible middle is **6-12 reps**: heavy enough to build, not so heavy that technique falls apart.

Proximity to failure, technique, range of motion

For a set to do something, it has to be hard, but you do not have to go all the way to failure, and as a beginner you should not. Work at **RIR 1-3** (RIR = how many reps you could still do; stop when 1-3 clean reps are left in the tank). Keep a **full range of motion** and clean technique. A short range and bouncing steal results and raise injury risk. A slightly lighter weight done perfectly beats a heavy mess.

Rest between sets

Don't rush from one set into the next. On the big compound lifts (squat, presses, pulls) take about **2-3 minutes**. A longer rest keeps performance and strength in the later sets and supports long-term gains [5]. On smaller isolation exercises, about 60-90 s is enough. Treat the rest as part of the session.

Exercise choice and structure

Build the base on **compound exercises** (squat, deadlift, presses, rows). They cover the most muscle in the least time and carry most of the progression. Add a few **isolation exercises** (biceps, triceps, shoulders, calves) to tidy up weaker areas. How often you train decides the structure: if you train **3 times a week, run full-body**. If you train **4 times a week, split upper and lower**. Both are written out below.

Before you start

On every exercise, stay at RIR 1-3. Stop when you could still do 1-3 clean extra reps. As a beginner, don't go to failure. Between sets of the big lifts, take **2-3 min**. On the smaller ones, 60-90 s [5]. Training at home? Instead of a barbell, a resistance band, a backpack with books, or water bottles is fine. The principle of progressive overload stays the same. You just add resistance over time [1].

> **Warm-up (5-8 min, every session):** 3-5 min of easy movement (walking, a bike, jumping rope), then move the joints (arm circles, unloaded squats, lunges, wall push-ups). Before the main lifts, do 1-2 light "feel-out" sets.

Option A: full-body, 3 times a week (Mon / Wed / Fri)

For 3 sessions a week this is the most effective choice: you hit each muscle group 3 times a week, so you get frequent practice and a decent weekly volume [2,3]. Leave at least one day off between sessions. Below are three different full-body days, so the exercises rotate.

Day 1: Monday

Mon: full-body A

Exercise	Sets × reps	Technique note
Squat (bodyweight / barbell / goblet)	**3 × 8–12**	Back braced, knees out, go as deep as you can control.
Push-up (knees / hands on a bench / full)	**3 × 6–12**	Body in one line, elbows slightly toward the ribs, lower slowly.
Bent-over row (dumbbells / band)	**3 × 8–12**	Back flat, pull the elbows back, squeeze the shoulder blades.
Romanian deadlift (light load)	**3 × 8–10**	Hips back, knees softly bent, you feel the back of the thighs.
Plank	**3 × 20–40 s**	Brace the stomach. Don't let the hips sag or pike up.

Day 2: Wednesday

Wed: full-body B

Exercise	Sets × reps	Technique note
Split squat, stationary (bodyweight / dumbbells)	**3 × 8–10 / leg**	Front knee over the ankle, torso upright.
Overhead press (dumbbells / band)	**3 × 8–12**	Brace the stomach, don't arch the back, elbows under the wrists.
Pulldown / high row (band / cable)	**3 × 8–12**	Pull the elbows down toward the ribs, shoulders down away from the ears.
Hip thrust / glute bridge	**3 × 10–15**	Lift the hips with the glutes, short pause and squeeze at the top.
Dead bug	**3 × 8–10 / side**	Low back pressed into the floor, slow and controlled.

Day 3: Friday

Fri: full-body C

Exercise	Sets × reps	Technique note
Goblet squat (dumbbell / kettlebell / backpack)	**3 × 8-12**	Load at the chest, chest up, sit down evenly.
Bench press / push-up with feet raised	**3 × 6-12**	Shoulder blades pulled together, smooth, no bouncing.
One-arm dumbbell row on a bench	**3 × 8-12 / side**	Back flat, pull the elbow back and up along the body.
Step-up	**3 × 8-10 / leg**	Whole foot on the step, drive through the heel, don't push off the back foot.
Side plank	**3 × 15-30 s / side**	Body in a line, hips up, shoulder blade braced.

Option B: upper/lower, 4 times a week

If you train 4 times a week, split the body into upper and lower and alternate them. You then meet each muscle group 2 times a week, a good frequency at the same volume or a bit more [2,3]. A typical week: Mon upper, Tue lower, Thu upper, Fri lower (or any 4 days with a gap between the same type of session). Take the exercises and the numbers from the tables above and sort them into the blocks below.

A **Upper block**

Bench press / push-up, overhead press, bent-over row, pulldown / high row, plus a short isolation (biceps, triceps). About 3-4 exercises × 3 sets.

B **Lower block**

Squat, deadlift / Romanian deadlift, lunges or step-ups, hip thrust, plus trunk (plank, dead bug). 3-4 exercises × 3 sets.

C **Frequency**

2 upper + 2 lower per week. On the second upper or lower day, change the exercises a little so you load the muscle from more angles.

D **Volume**

Target about 10-20 sets per muscle group per week [2]. Start at the low end and add only when you are progressing regularly.

How to progress (progressive overload)

Muscle grows when you regularly give it a little more work than last time [1,6]. You don't have to chase everything at once. Move one variable at a time.

1 Reps first
Keep the weight and add reps until you can hit the top of the range on every set.

2 Then add weight
When you hit the top (for example 3 × 12) in two sessions in a row, add a small step and go back to the bottom of the range.

3 How much to add
Lower body (squat, deadlift) about 2.5-5 kg. Upper body (presses, rows) about 1-2.5 kg. At home: a heavier backpack, a stronger band, a slower tempo.

4 Technique first
Never add weight at the cost of the movement. Sloppy technique slows progress and raises injury risk.

How often to add, and when to deload

- ✓ A realistic pace: a small step (a rep or the weight) about every 1-2 weeks on a given exercise. Progress is not a straight line forever.
- ✓ Start with 3 sets per exercise. Over a few weeks you can add a set on the main lifts. A weekly volume of about 10+ sets per muscle group is a good target for growth.
- ✓ **Deload** when you are worn out, performance stalls for 2 sessions in a row, joints ache, or you don't feel fresh. Typically every 6-8 weeks.
- ✓ A deload week = drop the weight by about 40-50%, or drop a set. Keep the technique and the movement the same. The body sheds fatigue and you come back stronger.

Recovery: protein and sleep

Training is the stimulus. Growth happens in recovery. Two things decide most of it. **Protein:** enough protein clearly increases the muscle and strength you get from resistance training. The target is about **1.6 g per kilogram of body weight per day**, spread through the day [8,9]. **Sleep:** without decent sleep the body does not get to repair what the session broke, and both performance and the will to work drop. The detail is in Protein and high-protein recipes and Sleep and recovery. Treat them as part of the training plan, not as a bonus.

Action step (how to progress): Hold the weight and add reps. Once you can do the **top of the range** on every set of an exercise (for example 3 × 12) in two sessions in a row, **add a small step of weight** and go back to the bottom of the range. Then climb again. That is progressive overload in practice. It is the main reason you will get stronger [1,6]. 💪

Practical tip: Keep a simple log: date, exercise, weight, sets and reps. Without the numbers you simply cannot tell whether you are moving. A note on your phone is

enough, or a piece of paper next to the table above.

© Martin Barna: client guide. Share only with the author's permission. martinbarna.cz/en

Sources: [1] Schoenfeld BJ. The mechanisms of muscle hypertrophy and their application to resistance training. J Strength Cond Res. 2010;24(10):2857-2872. [2] Schoenfeld BJ, Ogborn D, Krieger JW. Dose-response relationship between weekly resistance training volume and increases in muscle mass: a systematic review and meta-analysis. J Sports Sci. 2017;35(11):1073-1082. [3] Schoenfeld BJ, Ogborn D, Krieger JW. Effects of resistance training frequency on measures of muscle hypertrophy: a systematic review and meta-analysis. Sports Med. 2016;46(11):1689-1697. [4] Schoenfeld BJ, et al. Strength and hypertrophy adaptations between low- vs. high-load resistance training: a systematic review and meta-analysis. J Strength Cond Res. 2017;31(12):3508-3523. [5] Grgic J, et al. Effects of rest interval duration in resistance training on measures of muscular strength: a systematic review. Sports Med. 2018;48(1):137-151. [6] American College of Sports Medicine. Progression models in resistance training for healthy adults. Med Sci Sports Exerc. 2009;41(3):687-708. [7] Ralston GW, et al. The effect of weekly set volume on strength gain: a meta-analysis. Sports Med. 2017;47(12):2585-2601. [8] Morton RW, et al. Protein supplementation on resistance training-induced gains: meta-analysis. Br J Sports Med. 2018;52(6):376-384. [9] Helms ER, Aragon AA, Fitschen PJ. Evidence-based recommendations for natural bodybuilding: nutrition and supplementation. J Int Soc Sports Nutr. 2014;11:20. Always factor in your health. If something is wrong, talk to your doctor.